DIAdem Help

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DIAdem VIEW

DIAdem VIEW

Use the DIAdem VIEW panel to display your data as curves or surfaces, and also in tables, and to add text, graphics, or videos. The DIAdem VIEW function group includes predefined layouts, which you can use to display your data. You can also create your own worksheets and save them in a VIEW layout. You can organize the pages of a layout in the Manage Worksheets dialog box. Click the symbol **Transfer to REPORT** on the toolbar to create a report page from the current worksheet.

You can separate every worksheet into several areas and each area can use a different display type:

- Select Channel Tables to list your data in a table.
- Select 2D Axis System to display data as a curve.
- Select Bode to display a phase and an amplitude of a transfer function as superimposed curves.
- Select Polar to display your data as curves in a polar axis system.
- Select Orbit to display the rotation movement of a wave as an array of curves.
- Select Shaft Centerline to display the deflection of a rotating wave from the idle position as a curve.
- Select Bird's Eye View to display radar data - including the course of the road - lane, object list, and distances from a bird's eye view in DIAdem VIEW. You can synchronize this data with GPS data and objects in the video.
- Select Contour to display 3D data two-dimensionally in a contour display.
- Select Cascade to display data as a waterfall.

- Select Map to display geographic data, for example, the data of a GPS receiver, in a map section.
- Select Graphic to insert a picture in your layout.
- Select Video to insert a film of the test into your layout.
- Select Textbox to insert texts and variable contents into your layout.
- Select Dialog Box to insert a user dialog box into your layout.

You can drag and drop data channels from the Data Portal into axis systems, channel tables, and contour displays.

You can create, delete, and rename VIEW areas with the context menu. To move a VIEW area, click the blue title bar of the active area and drag it to the desired position. A small icon on the cursor indicates the position of the area after you release the mouse button. If you hold down the control key while dragging, DIAdem copies the area instead of moving it. To move or copy an area to another worksheet, move mouse cursor to the tab of the desired worksheet and wait a moment to activate the worksheet. To move or copy an area to a new worksheet, move the mouse pointer behind the last tab of the worksheets.

In axis systems you analyze the curves with Cursors and zoom sections. You can synchronize several axis systems in a worksheet over the reference axis. In the Legend of an axis system you specify the leading curve on which the cursor moves.

	Note The Transfer to Report function does not create every display type in DIAdem REPORT. DIAdem does not transfer display types for which DIAdem REPORT does not have equivalent display forms or transfers the display types only as a picture. For example, DIAdem REPORT displays only the last display of a video, a map, or a dialog box.

	Note If you use a script to modify the internal data in the Data Portal, you must call the Refresh method to update the layout in DIAdem VIEW.

Procedures

Block Operations in Channel Tables | Changing a Table Display | Configuring a Map

Display | Configuring a Text Box Display | Creating Areas | Cutting, Copying, and Pasting Data Blocks into Channel Tables | Deleting Channel Table Columns | Deleting Curve Sections and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying All Channels in One Channel Table | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Generating and Formatting Text | Interpolating Curve Points | Moving or Copying Areas | Partitioning Worksheets | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

Examples

Analyzing and Displaying Channels Section by Section | Automatic Display of Harmonic Frequencies | Background Segments in DIAdem VIEW | Calculating a Tangent to a Curve | Creating and Viewing Long Data Channels | Dynamic Display of Statistical Characteristic Values in DIAdem VIEW | Overlaying Objects in Videos | User Dialog Box in VIEW for the Calculation of the FFT and Harmonic Frequencies | Vibration Data Analysis with Parallel Processing | Viewing and Automatically Analyzing Data

DIAdem VIEW > Channel Table

Channel Table

Select **Display Type»Channel Table** in the context menu of a VIEW area to display channels numerically in a table.

Open the Display dialog box in the context menu of the channel table and specify the table data. You can specify whether DIAdem displays all the channels loaded in the Data Portal, all the channels of a group, or only selected channels, in the channel table. If you display all channels in the Data Portal or from a group, the channel table automatically displays new channels, for example, calculation results. DIAdem displays an empty column next to the table. If you enter values in this column, DIAdem automatically creates a new channel. If you enter values that exceed the maximum channel length, DIAdem automatically extends the channel.

If DIAdem displays only selected channels in the channel table, you can modify the

order of display in the **Display** dialog box, or you can drag and drop channels within the channel table. You can select the columns of a channel table and copy them into a different area using drag and drop, or move them by pressing the shift key.

Select Select Properties from the context menu to specify which properties DIAdem displays in the channel heading. If you drag and drop channel properties from the Data Portal into a channel table in DIAdem VIEW, DIAdem adds the contents of the properties automatically as a new row to the table header.

Use Synchronize with Cursor in the context menu of the channel table to specify that DIAdem highlights the rows in the channel table which correspond to curve points you selected with the cursor in the curve in a different area.

The context menu in the channel contents has various functions for editing the data. For example, use Block Operations to select, copy, or insert data blocks over several columns. Select Create New Channel to create new channels. Select Generate to generate equidistant values. You also can edit each table column individually. Use **Go To** to position the cursor at a specific point in the channel table. For example, use **Go To**»**Cursor Position** to position the cursor on the table column which corresponds to the position of the current curve cursor, the maximum values cursor, or the minimum values cursor in the axis system of the same layout page.

Use Lock Columns from the context menu of a selected column to freeze this column and all columns on the left of it when scrolling through the table. Select a column and click **Unlock Columns** in the context menu to disable the lock.

Use **Print Table** to print the table or the selected range of the table.

Procedures

Block Operations in Channel Tables | Changing a Table Display | Cutting, Copying, and Pasting Data Blocks into Channel Tables | Deleting Channel Table Columns | Displaying All Channels in One Channel Table

DIAdem VIEW > 2D Axis System

2D Axis System

Select **Display Type**»**2D Axis System** in the context menu of a VIEW area to display your channels as curves. You need a minimum of one and a maximum of two channels in order to define a curve.

Select the sub-menus of **X-Axis** or **Y-Axis** from the context menu to display more curves in one or several axis systems with absolute values or percent values, or to change between linear or logarithmic axes. You also can access these functions via the symbols on the axis system toolbar. Click the button **Select Curve** on the axis system of the toolbar to select a curve in the axis system. In the context menu you can delete the selected curve or you can define it as the Leading curve.

Open the Display dialog box in the context menu to specify the data channels for the curve display, to define Constants and Background Segments, and to change the Cursor Synchronization.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every 2D axis system.

In the context menu you can show hidden curves, delete selected curves, or delete all curves in the axis system. Select **Go To**»**Cursor Position** from the context menu of the channel table on the same layout page in order to position the cursor in the table row which corresponds with the current cursor position in the axis system.

You can play waveform channels acoustically. Click the loudspeaker symbol on the toolbar and then **Start** or **Repeat Play**. DIAdem plays the waveform channel synchronous to the cursor. Specify the play speed in the Cursor Parameters dialog box. If you enable the minimum or the maximum cursor, there is no sound. The Channel properties **Waveform x-Unit** and **Waveform x-step width** must be a time unit and the **Waveform x-step width** must be greater than 5,208333 µs in order to play waveform channels. The channel must be completely accessible in the computer memory and not just registered. Some sound cards cannot play all frequencies. Most sound cards support the frequencies 16 kHz, 22.05 kHz, 44.1 kHz, 48 kHz.

Click the loudspeaker symbol to switch off the sound. You also can enable and disable the sound if you select **Sound On** from the context menu of a 2D axis system. Use the Sound for CurveChart2D property to enable or to disable the sound with a script.

Procedures

Configuring a Map Display | Configuring a Text Box Display | Deleting Curve Sections and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Interpolating Curve Points | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

Examples

Analyzing and Displaying Channels Section by Section | Automatic Display of Harmonic Frequencies | Background Segments in DIAdem VIEW | Calculating a Tangent to a Curve | Creating and Viewing Long Data Channels | Dynamic Display of Statistical Characteristic Values in DIAdem VIEW | Overlaying Objects in Videos | User Dialog Box in VIEW for the Calculation of the FFT and Harmonic Frequencies | Vibration Data Analysis with Parallel Processing | Viewing and Automatically Analyzing Data

(2D Axis System | Cursor Types) > Zoom Functions

Zoom Functions

Use the zooming function to zoom into sections of the active area in order to examine sections of the curve in details. DIAdem VIEW offers various zoom types in the context menu of the axis systems or on the axis system toolbar. DIAdem scales the area according to the zoom. The video and graphics displays provide zoom functions in the context menu.

	Mouse wheel	Use the mouse wheel to enlarge and reduce the active area centrically. If you press <Ctrl> while you zoom, DIAdem zooms in larger steps. In the **Channel Table** you can scroll with the mouse wheel through the displayed table columns. To scroll linewise through several table columns, Press <Shift> at the same time. To zoom the channel table, press <Ctrl> simultaneously.
	Zoom	Click **Zoom In** on the toolbar of an axis system in order to enlarge the display in the

	In	active area. You can also use the plus key.
	Zoom Out	Click **Zoom Out** on the toolbar of an axis system in order to reduce the display in the active area. You can also use the minus key.
	Right Mouse Button	Press the right mouse button and drag open a band or a frame in order to enlarge a section in the active area. DIAdem fills the area with the selected section.
	Band Zoom	Click **Band Zoom** on the axis system toolbar and press the left mouse button to select and enlarge an axis section. You can repeat the zoom so that you can examine ever smaller areas.
	Frame Zoom	Click **Frame Zoom** on the axis system toolbar and press the left mouse button to select and enlarge a section. You can repeat the zoom so that you can examine ever smaller areas.
	Move	Click **Move** on the axis system toolbar and press the left mouse button to move the enlarged display.
	Scroll in Cursor Range	Select **Band Cursor** or **Frame Cursor** in the axis system and click **Sroll in Cursor Area** on the toolbar. To move along the enlarged curve, click and drag the cursor horizontally. If there are further axis systems with the same value range on the worksheet, the cursor also moves in these axis systems.
	Zoom Off	This function on the axis system toolbar switches off every zoom function. DIAdem restores the original curve.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

DIAdem VIEW > Bode Axis System

Bode Axis System

Select **Display Type**»**Bode** in the context menu of a VIEW area to display the phase shift and the amplitude amplification of a complex transfer function as super-imposed curves. You need a minimum of two and a maximum of three channels in order to define a curve.

Use the submenus of the **Amplitude** and **X-Axis** in the context menu or the toolbar of

the axis system, in order to scale the y-axis of the amplitude display and the joint x-axis linear of logarithmic.

Open the Display dialog box in the context menu to specify the data channels for the curve display and to change the Cursor Synchronization.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

In the context menu you can show hidden curves, delete selected curves, or delete all curves in the axis system.

Procedures

Configuring a Map Display | Configuring a Text Box Display | Deleting Curve Sections and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Interpolating Curve Points | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

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DIAdem VIEW > Polar Axis System

Polar Axis System

Select **Display Type**»**Polar** in the context menu of a VIEW area to display your channels as curves in polar axis systems. You need a minimum of one and a maximum of two channels in order to define a polar curve.

DIAdem displays the polar axis system as a full circle. You can change the angle direction and also the position of the zero-point for the polar axis system. DIAdem displays in the top left of the area whether the angle direction is clockwise or counter clockwise.

Open the Display dialog box in the context menu to specify the data channels for the curve display and to change the Cursor Synchronization.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

In the context menu you can show hidden curves, delete selected curves, or delete all curves in the axis system.

Procedures

Configuring a Map Display | Configuring a Text Box Display | Deleting Curve Sections and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Interpolating Curve Points | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

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Orbit

Select **Display Type**»**Orbit** in the context menu of a VIEW area to display the movements of a rotating shaft as a curve. To define a curve, you must specify two channels.

DIAdem always displays the Orbit axis system quadratically and both axes scaled the same because the values of the two sensors always have the same unit. In the axis system DIAdem displays the rotation direction of the shaft and also the position of the two sensors. If you specify a maximum bearing clearance, DIAdem displays this limit as a curve or an ellipse.

Open the Display dialog box in the context to specify the data channels for the curve display, to change the Cursor Synchronization, and to specify the shaft settings.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

In the context menu you can show hidden curves, delete selected curves, or delete all curves in the axis system.

Procedures

Configuring a Map Display | Configuring a Text Box Display | Deleting Curve Sections and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Interpolating Curve Points | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

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DIAdem VIEW > Shaft Centerline

Shaft Centerline

Select **Display Type**»**Shaft Centerline** in the context menu of a VIEW area to display the deflection of a rotating shaft from the idle position as a curve. To define a curve, you must specify two channels.

DIAdem displays the Shaft Centerline axis system quadratically and both axes scaled the same because the values of the two sensors always have the same unit. In the axis system DIAdem displays the rotation direction of the shaft and also the position of the two sensors. If you specify a maximum bearing clearance, DIAdem displays this limit as a curve or an ellipse.

Open the Display dialog box in the context to specify the data channels for the curve display, to change the Cursor Synchronization, and to specify the shaft settings.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

In the context menu you can show hidden curves, delete selected curves, or delete all curves in the axis system.

Procedures

Configuring a Map Display | Configuring a Text Box Display | Deleting Curve Sections

and Interpolating over Specified Points | Displaying 2D Axis Systems with Several Y-Axes | Displaying Images in DIAdem VIEW | Displaying Properties in the Legend of a Curve | Displaying the Curve Coordinates | Displaying Videos and Data Synchronously | Finding Values in Channel Tables | Interpolating Curve Points | Setting the Leading Curve | Transferring VIEW Layouts to DIAdem REPORT | Viewing Data as Curves | Zooming and Scrolling Curve Areas

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DIAdem VIEW > Bird's Eye View

Bird's Eye View

Select **Display Type**»**Bird's Eye View** from the context menu of a VIEW area to display radar data - including the course of the road - lane, object list, and distances from a bird's eye view in DIAdem VIEW. You can synchronize this data with GPS data and the detected objects in the video.

Use **Play**, **Repeat Play**, or **Pause** in the toolbar, and the functions for fast forward or fast reverse and step-by-step forward or step-by-step reverse in the Position Cursor dialog box, to control the display. If your layout contains a bird’s eye view and an axis system, and you move the cursor across the axis system, DIAdem runs a synchronous playback of the bird’s eye view and the cursor position.

Examples

Bird's Eye View Display | Planetary Motion

DIAdem VIEW > Contour

Contour

Select the **Display Type**»**Contour** in the context menu of a VIEW area to display 3D data two-dimensionally in a surface. The contour display shows the third dimension of the data as a color value or as isolines. You can use a contour display, for example, to display and to analyze characteristic diagrams in DIAdem VIEW.

Open the Display dialog box in the context menu to specify the data channels for the contour display and to define several isolines. You cannot create isolines until you have defined a contour. This means you delete all associated isolines together with a contour.

Click the **Display Dynamic Isoline** button on the toolbar of the contour display to display an isoline at the cursor position. DIAdem displays the dynamic isoline unlabeled in black. If you move the cursor, DIAdem automatically refreshes the displayed isoline. The **Display Dynamic Isoline** function is also in the context menu.

Click the **Create Isoline at Cursor Position** button on the toolbar of the contour display to add an isoline to the contour display at the current cursor position. DIAdem calculates the isoline for the current z-value of the cursor, saves the result channels in the Data Portal default group, and then displays the result in the contour display.

Click the **Create and Refresh Channels for Intersection Curves** button on the contour display toolbar to specify intersections for the displayed 3D data. DIAdem determines sections that are parallel to the x-axis and the y-axis for the current cursor position. DIAdem saves the associated channels in a separate channel group of the Data Portal. You can display the calculated sections in 2D axis systems and synchronize them with the contour display. If you move the cursor in the contour, DIAdem automatically refreshes the calculated sections. The **Create and Refresh Channels for Intersection Curves** function is also in the context menu.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. Use flags to copy or to delete single points or parts of a curve. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

Examples

Contour Display in DIAdem VIEW | Various Display Modes in DIAdem VIEW

DIAdem VIEW > Cascade

Cascade

Select **Display Type**»**Cascade** in the context menu of a VIEW area to display data as a waterfall display.

Open the Display dialog box in the context menu to specify the data channels for the cascade display. You can also define several 3D curves.

You create order lines based on an order channel. Afterwards, you create the order lines in the dialog box or use the functions on the toolbar of the cascade display. Click **Additional Order Lines** in the dialog box and create one or more static order lines by defining the orders. Click **Display dynamic order lines** on the cascade display toolbar to show the order lines at the current cursor position. If you click **Generate order line at cursor position**, a static order line displays in the cascade. DIAdem calculates the order line for the current z-value of the cursor, saves the result channels in the Data Portal default group, and enters the new order line into the curve list of the cascade.

Use **Generate and refresh dynamic orderlines** to show the order line curve in a dynamic display in a 2D axis system. DIAdem stores the x-channel and the y-channel of the order line in the Data Portal. Use these channels to define the curve in the 2D axis system. If you move the cursor over the cascade, DIAdem displays the current order line in the cascade. At the same time, DIAdem recalculates the values of the two channels and displays the refreshed order line curve in the 2D axis system.

Select Cursor from the context menu or on the toolbar in DIAdem VIEW to measure a curve and to zoom curve sections. You can set the leading curve in the Legend that DIAdem displays next to every axis system.

Examples

Cascade Display

Map

Select **Display Type**»**Map** in the context menu of a VIEW area to display geographical data, for example, the data of a GPS receiver, in a map section.

Select Display from the context menu of the map display to specify which data DIAdem displays and how DIAdem displays the data. To display maps, you can access maps from OpenStreetMap or from a user-defined map server.

The OpenStreetMap project collects geographical data in a database in order to create maps. OpenStreetMap data is free of licensing costs and you can use, edit, and customize the data according to your needs. Visit http://www.openstreetmap.org for further information on OpenStreetMap. You can enable a cache for the **OpenStreetMap** map type, so that DIAdem saves all displayed map sections as PNG files on the computer.

If you select the **Cache** map source, you can use the map sections without an internet connection. You can also use the cache mode to speed up the map display. If both panels use the same cache path, you can use the buffered maps also for an online analysis in DIAdem DAC.

Use the **User-defined** map type to access maps on a user-defined server.

You can zoom, reduce, and move the displayed map section. If your layout contains a map display and an axis system, and you move the cursor across the axis system, DIAdem displays the map section synchronously to the cursor. As a default, DIAdem moves the map section in such a way that the position marker is in the middle of the map section. The Synchronization of the position cursor in the map section with the cursor in the axis system is executed over the longitude and latitude channels. You can also use the synchronization track to synchronize. To do so, DIAdem needs a monotonic increasing synchronization channel with the same sampling rate as the longitude channel and the latitude channel, for example, a channel with the time stamps of the position data. Assign a cursor dimension to the synchronization track if, for example, you want to display the corresponding map section when you play a video.

The map display in DIAdem VIEW uses the Standard World Geodetic System 1984 (WGS 84) as a reference system in cartography. This system describes an idealized earth model, which takes into account all irregularities of the earth's shape.

Examples

Bird's Eye View Display | Map Display in DIAdem VIEW | Synchronizing Maps, Videos, and Measurement Data

DIAdem VIEW > Map > User-Defined Map Source

User-Defined Map Source

Define your own map sources for the map display in DIAdem VIEW by using the following variables to determine the map source, the maximum magnification, and the map provider in a script.

If you register the script in DIAdem as a user command at **Settings»Extensions»User commands** and save the desktop file, the additional map source will be permanently available in DIAdem. In the Map Display of DIAdem VIEW, select **User-defined** as the map source to access this map material.

Parameter

VIEWMapUserURL	Specifies the web address of the map source according to the OpenLayers syntax. You can also define several map servers from the same provider in one array. Variant
VIEWMapUserMaxZoom	Specifies the maximum zoom level of the map server. Longinteger 1 <= VIEWMapUserMaxZoom <= 30 Access: Read/Write
VIEWMapUserAttribution	Specifies the copyright of the map provider.

	String Access: Read/Write

Example

The following example defines a map source on a server in the local network.

VBScript	Python

Copy script

```
VIEWMapUserURL="http://192.168.228.129/tile/{z}/{x}/{y}.png"
VIEWMapUsermaxZoom=20
VIEWMapUserAttribution="(c) OpenStreetMap (and) contributors, CC-BY-SA"
```

The following example defines **OpenCycleMap** as the map source. The map source requires an API key, which you transfer as a parameter. To find out how to obtain the API key, visit the map provider's website.

VBScript	Python

Copy script

```
VIEWMapUserURL="https://tile.thunderforest.com/cycle/{z}/{x}/{y}.png?apikey="
VIEWMapUsermaxZoom=22
VIEWMapUserAttribution="Maps © Thunderforest, Data © OpenStreetMap contributors"
```

Examples

Bird's Eye View Display | Map Display in DIAdem VIEW | Synchronizing Maps, Videos,

and Measurement Data

DIAdem VIEW > Graphic

Graphic

To insert an image into the layout, select **Display Type**»**Graphic** from the context menu of a VIEW area. Double-click an image area to select a new graphic.

Use the Zoom from the context menu to zoom in and out of a graphic in a presentation. Select **Keep Aspect Ratio** from the context menu to display the graphic without distortions.

DIAdem VIEW > Video

Video

Select **Display Type**»**Video** from the shortcut menu of a VIEW area to insert a video of a test procedure into the layout. In DIAdem VIEW, you can play back videos in MPEG, MPG, WMV, and AVI format. The videos must be freely positionable, so that DIAdem can display each frame in the videos. Refer to Requirements for Playing Videos in DIAdem VIEW for further information.

Use **Play**, **Repeat Play**, or **Pause** in the toolbar, and the functions for fast forward or fast reverse and step-by-step forward or step-by-step reverse in the Position Cursor dialog box, to control the video. If your layout contains a video and an axis system, and you move the cursor across the axis system, DIAdem plays the video synchronously to the cursor. Select Display from the context menu to specify the speed and the duration of the video.

Before playing the video, you can select Zoom from the context menu to enlarge the video display. You can also enable **Sound On** in the context menu to play the video with sound. In a script you use the property Sound for CurveChart2D to do so.

Examples

3D Interpolation | Bird's Eye View Display | Overlaying Objects in Videos | Planetary Motion | Recording Video and Measurement Data | Synchronizing Maps, Videos, and Measurement Data | Synchronizing Videos and Data | Synchronizing Videos and Data | Synchronizing Videos and Waveform Data | Synchronizing Videos and Waveform Data | Various Display Modes in DIAdem VIEW

DIAdem VIEW > Video > Requirements for Playing Videos in DIAdem VIEW

Requirements for Playing Videos in DIAdem VIEW

To be able to play a video you must install the codec with which the video file was coded. Additionally, the software used for video playback must support the format of the video file.

Codecs

Codecs are used for coding and decoding video signals. The coding, or encoding, usually includes a compression for reducing the data rate required for the transmission. The storage space required for storage in a file also decreases greatly. For the video playback the encoded video signals must be decoded with the same codec. Codecs with a higher compression rate usually require more computing power when encoding and playing videos.

Earlier codecs use a process similar to the JPEG compression to compress each frame in the video separately from the other frames. Newer methods additionally consider the fact that the differences between successive frames are very small. This method is called interframe compression and combines key frames and delta frames in order to increase the compression of video files. Interframe compression is used especially for video streaming. Codecs without interframe compression, for example, the DV codec, are still important for video cutting and as an exchange format.

In an interframe compression the key frame contains all information required to display this frame. The delta frame, however, requires data from other frames in order to display. All videos contain at least one key frame at the beginning. Additional key frames improve the continuity of the video but also increase the file size. The more key

frames a video contains, the easier the positioning within the video is during playback.

The encoded video signals that a codec generates are saved in a file. Codecs do not always have the same name as the file format they create. Therefore it is important to differentiate between codecs on the one hand and file formats or container formats on the other hand. You can save various video and audio tracks that were encoded with different codecs in one container format such as AVI.

Container Formats

In software technology, a file format that can contain different file formats is called a container. A container format only defines the type and structure of how to save the contents. Containers facilitate, for example, the storage of audio and video tracks in one file. Often the container formats AVI, MPEG-1, MPEG-2, or MPEG-4, are used for videos. An AVI file can combine contents such as bitmaps, audio and video files, or RTF texts that were encoded with different methods. The content type is saved with a Four Character Code (FourCC). This four character long string in the header of the AVI file specifies which codec to use for decoding the data.

File Formats

Only few file formats, such as the WMV format (Windows Media Video) have established themselves. The WMV format is a file format defined by Microsoft contains video and audio streams coded by default with the Windows Media Video 9 Codec (WMV9). Video files can also be generated with Windows Media Video 7 Codec or Windows Media Video 8 Codec (WMV7, WMV8). The Microsoft Windows media platform provides the necessary video and audio codecs.

Videos in DIAdem

When playing videos in DIAdem VIEW, the main focus is on the synchronization of measurement data and video signals. To synchronize the time information of the various data sources, DIAdem must position the video to certain time points. This is the only way to synchronize the video display when you scroll with the cursor through the measurement data. Positioning the video is always easier when the codec does not use interframe compression. If you reduce the distance between the key frames in the settings of the codec with interframe compression, you can improve video playback in

DIAdem VIEW. The video wobbles less during playback and the wobbling starts later within the video.

Limitation 1: If the playback speed is not equal to 1 with unchanged frame rate, the video may jerk. The jerking increases as the playing time progresses. For example, the jerking may start at 10 seconds, become stronger at 20 seconds, and at 40 seconds the video might stop.

Restriction 2: the same as restriction 1. However, the video might jerk less as long as the playback speed remains in the range of 0.5 to 2.0.

DIAdem Cannot Play a Video

Depending on your operating system, you may need to download the MPEG-2 video extension from the Microsoft Store to view a video.

By default the multimedia functionality on the Windows Server operating system (Windows Server 2008R2 and later) is not installed. In order to install them, the "Desktop Experience" feature must be enabled. After you executed this, you can try again to open a video file.

Since DIAdem 2014, DIAdem has been ablt to read out and display the following information about a video:

- Codec-ID – Identifies the codec through a unique code.
- FourCC – The Four Character Code provides the type for the contents.

If you cannot play a video in DIAdem, you usually can open the video in the Windows Media Player. In this case, the Windows Media Player tries to install the necessary codec. If the Windows Media Player does not install the missing codec, execute a search with the information on FourCC and then on the codec ID in order to find the missing codec. The search for the FourCC code should lead to a quick result.

Use special software such as the free GSpot (`www.headbands.com/gspot`) to read out which codec you need for playing your video file if DIAdem does not return any additional information. A quick internet search usually provides the required codec. With the GSpot software you also can check which codecs are installed on the computer and which codec belongs to which Four Character Code (FourCC).

In most cases, an active audio device is required on your computer for DIAdem to play a video.

Error message "Codec does not support accurate synchronization": DIAdem cannot play the video file because the necessary positioning information is missing. Encode the video file with a different codec.

Examples

3D Interpolation | Bird's Eye View Display | Overlaying Objects in Videos | Planetary Motion | Recording Video and Measurement Data | Synchronizing Maps, Videos, and Measurement Data | Synchronizing Videos and Data | Synchronizing Videos and Data | Synchronizing Videos and Waveform Data | Synchronizing Videos and Waveform Data | Various Display Modes in DIAdem VIEW

DIAdem VIEW > Textbox

Textbox

Select **Display Type**»**Textbox** in the context menu of a VIEW area to display text and variable contents.

Click the textbox to enter and edit text. If you enter text that includes DIAdem variables and formulas, enclose the variables or formulas in @ characters. Click **Refresh** in the context menu to read the current variable values. Refer to Using Variables in Text for more information about displaying variable contents in text. If you drag and drop properties from the Data Portal into a text box in DIAdem VIEW, DIAdem adds the contents of the properties automatically as text.

Select Display from the context menu to specify the appearance of the text and to specify whether DIAdem displays scroll bars in the textbox.

Procedures

Generating and Formatting Text

DIAdem VIEW > Dialog Box

Dialog Box

Select **Display Type**»**Dialog Box** in the context menu of a VIEW area to insert user dialog boxes in DIAdem VIEW.

Use user dialog boxes to enter and display settings or to start functions. Select **Edit Dialog Box** from the context menu to open the dialog editor. Use the DIAdem Dialog Editor in DIAdem Script to create user dialog boxes that you save in a dialog file with the filename extension `*.sud`. In a dialog box file, you can save several dialog boxes, so that all the dialog boxes of a project are in the same SUD file.

User dialog boxes comprise controls such as buttons, entry and selection fields, graphics, and text. You can place these controls anywhere in the dialog box and design them as you like. You can use events to connect the elements in the user dialog box with logical operations. To display a user dialog box in DIAdem VIEW the size of this user dialog box must be modifiable and must contain a ViewConnector control. The **ViewConnector** control is not visible at runtime.

Select **Reload Dialog Box** from the context menu to refresh the display of an edited dialog box.

In scripts you use the VIEW object Dialog to access the dialog box area in DIAdem VIEW during runtime. Use the VIEW object Dialog, which contains properties, methods, and events that correspond to the user dialog box object Dialog, to edit the user dialog box displayed in this area.

Examples

Contour Display in DIAdem VIEW | Various Display Modes in DIAdem VIEW

DIAdem VIEW > Cursor

Cursor

Use the cursors **Crosshair Cursor**, **Band Cursor**, or **Frame Cursor** to measure one or more curves. You can click from one cursor type to another on the DIAdem VIEW

toolbar.

You can use the mouse or the arrow keys to move the cursor in the enabled axis system. The cursor runs parallel in the other axis systems if the range of values is the same. You can observe the coordinates of the cursors in the tooltip, the status bar, and the legend.

Cursor Types		The cursor mode determines the appearance of the cursor.
	Crosshair Cursor	The **Crosshair Cursor** consists of a vertical line and a horizontal line, each of which you can move separately. DIAdem indicates the cursor intersection point on the curve with a small square. This intersection point defines the x-cursor and y-cursor coordinates, which DIAdem displays in the coordinate window as x-Cursor1 and y-Cursor1. If you idle the mouse over one of the cursor lines, the cursor changes to a horizontal or vertical arrow. Click and drag to move the cursor line vertically or horizontally. DIAdem refreshes the coordinates in the coordinates window.
	Band Cursor	The **Band Cursor** consists of two parallel, vertical lines that define a curve section with a dx width. DIAdem uses the x-Cursor1 coordinate for the x-position marked by a small red square, and for the x-Cursor2 coordinate the x-position of the other line. If you idle the cursor over one of the lines, the cursor changes into two crossed double arrows. You can then click and drag the entire band along the curve.
	Frame Cursor	The **Frame Cursor** is a rectangular area framed by two horizontal and two vertical lines. The intersection point of two lines marked by a small square specifies the x-Cursor1 and y-Cursor1 coordinates. The x-Cursor2 and y-Cursor2 coordinates correspond to the intersection point diagonally across. The dx-cursor and the dy-cursor represent the distances between the horizontal and vertical lines and specify the frame height and width. The frame cursor works the same way as the band cursor. You also can move the frame cursor in vertical direction.
Cursor Modes		The cursor mode determines the function of the cursor.
	Free Cursor	The default cursor in DIAdem VIEW is the **Free Cursor**, which you can move freely in the active axis system. To measure the curve, select the curve cursor, the maximum cursor, or the minimum cursor.

	Curve Cursor	Use the **Curve Cursor** to move the cursor along the curve. You can display the coordinates of each point on the curve.
	Maximum Values Cursor	Use the **Maximum Cursor** to trace the local maximum values on the curve.
	Minimum Values Cursor	Use the **Minimum Cursor** to trace the local minimum values on the curve.
Analysis Cursor		The analysis cursor specifies the number of vertical cursor lines in a 2D axis system. DIAdem marks the main cursor with a small red square.
	Normal Cursor	In 2D axis systems, the **Normal Cursor** is the default cursor. It displays a main cursor which it uses for measurements.
	Harmonic Cursor	Select the **Harmonic Cursor** on the toolbar of a 2D axis system to measure intervals that are the same width as a frequency analysis. The harmonic cursor has several cursor lines to the right of the main cursor. These secondary cursors behave dynamically. The distance of the cursor lines to each other corresponds to the distance of the main cursor to the axis origin. As a result, the distances increase when the distance of the main cursor to the axis origin increases. Left-click the cursor to move the cursor line in horizontal direction. This also changes all cursor distances.
	Sideband Cursor	Select the **Sideband Cursor** on the toolbar of the 2D axis system in order to inspect fixed intervals on the left and on the right side of the main cursor. The sideband cursor has several additional cursor lines to the right and left of the main cursor. The distance between the individual cursor lines is identical and remains constant when the cursor is moved. Left-click the cursor to move the cursor line in horizontal direction. This also changes all cursor distances.
Cursor Coordinates		DIAdem displays the current coordinates in the tooltip of the cursor.
	Coordinate Window	Click **Coordinates** on the toolbar of DIAdem VIEW to open and close the coordinates display. DIAdem displays the cursor coordinates in a separate window.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

(2D Axis System | Cursor Types) > Zoom Functions

Zoom Functions

Use the zooming function to zoom into sections of the active area in order to examine sections of the curve in details. DIAdem VIEW offers various zoom types in the context menu of the axis systems or on the axis system toolbar. DIAdem scales the area according to the zoom. The video and graphics displays provide zoom functions in the context menu.

	Mouse wheel	Use the mouse wheel to enlarge and reduce the active area centrically. If you press <Ctrl> while you zoom, DIAdem zooms in larger steps. In the **Channel Table** you can scroll with the mouse wheel through the displayed table columns. To scroll linewise through several table columns, Press <Shift> at the same time. To zoom the channel table, press <Ctrl> simultaneously.
	Zoom In	Click **Zoom In** on the toolbar of an axis system in order to enlarge the display in the active area. You can also use the plus key.
	Zoom Out	Click **Zoom Out** on the toolbar of an axis system in order to reduce the display in the active area. You can also use the minus key.
	Right Mouse Button	Press the right mouse button and drag open a band or a frame in order to enlarge a section in the active area. DIAdem fills the area with the selected section.
	Band Zoom	Click **Band Zoom** on the axis system toolbar and press the left mouse button to select and enlarge an axis section. You can repeat the zoom so that you can examine ever smaller areas.
	Frame Zoom	Click **Frame Zoom** on the axis system toolbar and press the left mouse button to select and enlarge a section. You can repeat the zoom so that you can examine ever smaller areas.
	Move	Click **Move** on the axis system toolbar and press the left mouse button to move the enlarged display.

	Scroll in Cursor Range	Select **Band Cursor** or **Frame Cursor** in the axis system and click **Sroll in Cursor Area** on the toolbar. To move along the enlarged curve, click and drag the cursor horizontally. If there are further axis systems with the same value range on the worksheet, the cursor also moves in these axis systems.
	Zoom Off	This function on the axis system toolbar switches off every zoom function. DIAdem restores the original curve.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

DIAdem VIEW > Cursor Types > Flag Functions

Flag Functions

Use the flag functions to edit curves in DIAdem VIEW. Use flags to select curve points in order to copy, delete, or replace these curve points.

	Set Flags	Use the Crosshair Cursor to specify individual curve points and use the Band or Frame Cursor to specify curve sections. Click **Set Flags** on the axis system toolbar to select the curve points. If the axis system contains more than one curve, DIAdem sets flags only on the leading curve. To set flags on the specified section for all the curves, press the <Shift> key and click **Set Flags**. You can select several curve points which are not adjoined and also several curve sections. To do so, move the cursor over the curve and use **Set Flags** to select curve points.
	Flags: Copy Data Points	Use this function to copy the curve points selected with flags into new data channels.
	Delete Data Points	Use this function to delete the curve points selected with flags. Use **Settings»DIAdem Settings»VIEW** to specify whether DIAdem removes the deleted curve points from the data channel or replaces them with NoValues.
	Flags: Interpolate NoValues	Use this function to interpolate the curve sections specified by flags. The Settings in **Settings»DIAdem Settings»VIEW** determine whether the interpolation occurs through lines or splines.

	Set Data Point and Flag	Use this function to create new curve points. To do so, select the crosshair cursor and position the cursor on the new curve point. Click **Set Data Point and Flag**. DIAdem creates a new curve point and integrates it into the existing curve.
	Remove Flags	Use the Crosshair Cursor to specify one or more selected curve points and use the Band or Frame Cursor to specify curve sections. Click **Remove Flags** on the toolbar of the axis system. If the axis system contains more than one curve, DIAdem only deletes the flags on the leading curve. To delete the flags from all the curves in the selected area, press the <Shift> key and click **Remove Flags**.
	Remove Flags from All Data	Use this function on the DIAdem VIEW toolbar to delete all the flags of all the channels in all axis systems on the current page.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

DIAdem VIEW > Cursor Types > Cursor Synchronization

Cursor Synchronization

If you display the data as curves or as contours in different areas in DIAdem VIEW, DIAdem displays the cursors in these areas at the same cursor position with the same x-coordinates and y-coordinates. If the value ranges are the same and you move the cursor in an area, the cursors in the other areas follow passively.

This behavior results from the cursor dimensions X, Y, and Z that DIAdem creates automatically and assigns to all areas. For 2D axis systems and contour displays DIAdem assigns the cursor dimension X to the x-axis (abscissa) and assigns the cursor dimension Y to the y-axis (ordinate). DIAdem uses the cursor dimension Z for the heights of the contour displays. By default DIAdem uses the cursor dimension X for the time axis when DIAdem plays videos. If you move the cursor in the x-direction in an area, DIAdem writes the x-coordinate to the cursor dimension X. All other areas that also use the cursor dimension X for the cursor, read out the new x-coordinate and position the cursor accordingly. The same applies if the cursor moves in y-direction.

Axes	Default reference axes

	Abscissa: 2D, Orbit, Shaft Centerline	X
	Ordinate: 2D, Orbit, Shaft Centerline	Y
	Abscissa: Contour	X
	Ordinate: Contour	Y
	Ordinate: Altitude values	Z
	Abscissa: Bode	X
	Phase: Bode	Y
	Amplitude: Bode	Y2
	Angle: Polar	X
	Radius: Polar	Y

If the displayed data is not directly physically related, change the default cursor synchronization. For example, if you want to display noise data in one area and frequency spectrums in another area, use different cursor dimensions in the two areas. Select Settings»Cursor Parameters»Synchronization to define further cursor dimensions.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

DIAdem VIEW > Cursor Types > Legend

Legend

DIAdem displays a legend next to each axis system. To resize the legend area, click and drag the left edge of the area to the left.

	Curve Symbol	The legend displays a checkbox for each curve in the same color as the curve.
	Leading	If there are several curves in the axis system, click a curve symbol in order to

	Curve	specify the leading curve on which the curve cursor moves. You can also select a curve and click **Set Leading Curve** in the context menu.
☑	**Show Curves**	Enable the checkbox next to the colored symbol to specify whether DIAdem displays the curve. If you use this function, you can hide curves in the axis system without deleting the curve definitions.
	Select Curves	Click the area in front of a curve symbol to select the curve in the legend. Drag and drop the selected curves in order to copy them into a new area. If you press <Shift> at the same time, you move the curves out of the previous area into the new area.
	Invalid Curve	A curve definition is invalid if DIAdem cannot display the curve, for example, if the channel references are not valid. Use the context menu functions to delete invalid curves from the axis system.
	Properties	The legend always displays the name and the unit of each curve. Double-click the properties to open the Select Properties dialog box where you can specify further properties for the display in the legend. If you drag and drop properties from the Data Portal onto the legend, DIAdem adds the contents of this property to the legend automatically as the new row. If you drag the properties onto the 2D axis system, DIAdem creates a constant.

Further Settings

Cursor | Zoom Functions | Flag Functions | Cursor Synchronization | Legend

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